torch.autograd.functional.vhp#

https://pytorch.org/docs/stable/generated/torch.autograd.functional.vhp.html

Description:

Compute the dot product between vector v and Hessian of a given scalar function at a specified point. func (function) – a Python function that takes Tensor inputs and returns a Tensor with a single element. inputs (tuple of Tensors or Tensor) – inputs to the function func. v (tuple of Tensors or Tensor) – The vector for which the vector Hessian product is computed. Must be the same size as the input of func. This argument is optional when func's input contains a single element and (if it is not provided) will be set as a Tensor containing a single 1. create_graph (bool, optional) – If True, both the output and result will be computed in a differentiable way. Note that when strict is False, the result can not require gradients or be disconnected from the inputs. Defaults to False.

Function Signature

```
torch.autograd.functional.vhp(func, inputs, v=None,
create_graph=False, strict=False)[source]#

Parameters

Returns

tuple with:

Return type
```

Returns:

output (tuple)

Code Examples

```
>>> def pow_reducer(x):
...     return x.pow(3).sum()
>>> inputs = torch.rand(2, 2)
>>> v = torch.ones(2, 2)
>>> vhp(pow_reducer, inputs, v)
(tensor(0.5591),
 tensor([[1.0689, 1.2431],
         [3.0989, 4.4456]]))
>>> vhp(pow_reducer, inputs, v, create_graph=True)
(tensor(0.5591, grad_fn=<SumBackward0>),
 tensor([[1.0689, 1.2431],
         [3.0989, 4.4456]], grad_fn=<MulBackward0>))
>>> def pow_adder_reducer(x, y):
...     return (2 * x.pow(2) + 3 * y.pow(2)).sum()
>>> inputs = (torch.rand(2), torch.rand(2))
>>> v = (torch.zeros(2), torch.ones(2))
>>> vhp(pow_adder_reducer, inputs, v)
(tensor(4.8053),
 (tensor([0., 0.]),
  tensor([6., 6.])))
```

Detailed Documentation

Documentation

Compute the dot product between vector v and Hessian of a given scalar function at a specified point.

func (function) – a Python function that takes Tensor inputs and returns a Tensor with a single element.

inputs (tuple of Tensors or Tensor) – inputs to the function func.

v (tuple of Tensors or Tensor) – The vector for which the vector Hessian product is computed. Must be the same size as the input of func. This argument is optional when func's input contains a single element and (if it is not provided) will be set as a Tensor containing a single 1.

create_graph (bool, optional) – If True, both the output and result will be computed in a differentiable way. Note that when strict is False, the result can not require gradients or be disconnected from the inputs. Defaults to False.

strict (bool, optional) – If True, an error will be raised when we detect that there exists an input such that all the outputs are independent of it. If False, we return a Tensor of zeros as the vhp for said inputs, which is the expected mathematical value. Defaults to False.

func_output (tuple of Tensors or Tensor): output of func(inputs)

vhp (tuple of Tensors or Tensor): result of the dot product with the same shape as the inputs.